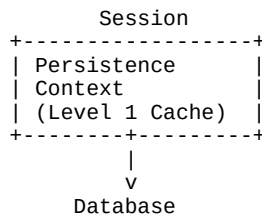


Chapter 6 - Hibernate Level 1 Cache (Detailed)

1. What is Level 1 Cache?

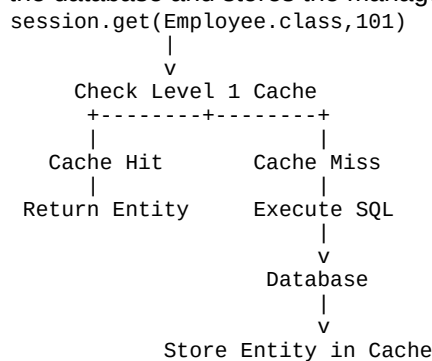
Level 1 Cache (First-Level Cache) is enabled by default in Hibernate. It is associated with a Session. Every Session has its own independent Level 1 Cache.

2. Internal Architecture



3. Entity Retrieval Flow

When an entity is requested, Hibernate first checks the Level 1 Cache. If the entity is not present, it queries the database and stores the managed entity in the cache.



4. Important Rules

- One Session = One Level 1 Cache.
- Two references to the same Session share the same cache.
- A new Session gets a new Level 1 Cache.
- The cache stores managed entity objects identified by their primary key.

5. Same Session Example

```
Session s = sessionFactory.openSession();

Employee e1 = s.get(Employee.class, 1); // SQL executes

Employee e2 = s.get(Employee.class, 1); // Returned from Level 1 Cache
```

6. New Session Example

```
Session s1 = sessionFactory.openSession();
s1.get(Employee.class, 1); // SQL
Session s2 = sessionFactory.openSession();
```

```
s2.get(Employee.class,1); // SQL again
```

7. Null Lookup

If `session.get(Employee.class,101)` returns null, Hibernate does not store null in the Level 1 Cache. Requesting the same entity with the same Session executes SQL again because only managed entity instances are cached.

8. clear(), evict() and close()

`clear()` removes all managed entities from the Session cache.

`evict(entity)` removes a single entity from the Session cache.

`close()` destroys the Session and its Level 1 Cache.

Operation	Effect on Level 1 Cache
<code>clear()</code>	Removes all cached entities
<code>evict(entity)</code>	Removes one cached entity
<code>close()</code>	Destroys Session and cache
New Session	Creates a new Level 1 Cache

9. Real-Time Scenario

In a service method, multiple calls to `get()` for the same primary key within one Session result in only one SQL query. This reduces database load and improves performance.

10. Interview Questions

1. What is Level 1 Cache? 2. Is it enabled by default? 3. Where is it stored? 4. Does it cache null values? 5. What happens after `clear()`, `evict()`, or `close()`? 6. Difference between Level 1 and Level 2 Cache?